Siddh Mehta

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AI/ML engineer who builds retrieval systems, LLM tools and production ML pipelines. Built and run ResearchIT, hybrid search and multi-interest recommendations over 1.85M arXiv papers. About two years across Airbus, Samsung R&D (PRISM), Fynd and Acheron.

EXPERIENCE

Airbus — ML / Data Engineer — Feb 2026 – Present
Bengaluru · Intern Feb–Aug 2026; full-time since Aug 2026

- Engineered PySpark ingestion and reconciliation pipelines processing **43M+** rows in Palantir Foundry, with lineage, automated health checks and validation gates feeding downstream models and reporting.
- Developed milestone-delay and mean-time-to-repair models (LightGBM, MiniLM-L12-v2 embeddings, SQL features, per-class thresholds) with scheduled retraining.
- Built contract-analytics and consumption-forecasting pipelines for financial provisioning, and a self-service application with dashboards for global operations and finance teams.

Samsung R&D Institute India — PRISM research program — Dec 2025 – Sep 2026
Remote

- Designed and benchmarked a two-stage document-image classifier (eight layout-aware classes) for constrained on-device inference, comparing CNN and transformer models on accuracy, latency and model size.

Fynd — AI/ML Engineering Intern — May 2025 – Jul 2025
Mumbai

- Built a modular Python/LangGraph multi-agent framework with reusable abstractions, cutting pipeline setup time **40%** and inference latency **60%**.
- Implemented hybrid dense–sparse RAG with BGE-M3; served inference through REST APIs with unit tests and A/B evaluation harnesses, improving on baselines by **8–15%** across three datasets.
- Fine-tuned a 1.5B-parameter Llama model with LoRA/QLoRA (PyTorch, PEFT), improving domain accuracy **9%**.

Acheron Software — Software Engineering Intern · earlier Data Science Intern — Jun 2024 – Apr 2025
Hyderabad / Remote · Software engineering intern Oct 2024 – Apr 2025; data science intern Jun – Aug 2024

- Shipped a LangChain Text-to-SQL application as an internal REST service (**89%** query accuracy) and a YOLOv8 asset-tagging pipeline with embedding similarity search (**92%** accuracy).
- Automated monitored data-processing and anomaly-detection workflows, saving about 20 minutes per upload cycle.

PROJECTS

ResearchIT — Personalized search and recommendations over 1.85M arXiv papers — 2025 – Present
Solo · siddhm11-researchit.hf.space · source

- Hybrid search over **1.85M** papers: BGE-M3 dense retrieval across three Qdrant shards plus SQLite FTS5, reciprocal rank fusion and a cross-encoder rerank; FastAPI/HTMX on Hugging Face Spaces with SQLite and Turso.
- Multi-interest feed: Ward clustering of saved papers into up to seven interests, importance-weighted quotas and per-interest MMR; traced and fixed a defect where one interest took 39 of 40 slots.
- Re-measured the reranker on production text before shipping it (MRR@10 **+7.7%** on 847 queries, 95% CI reported); restructured the FTS5 query to cut median lexical search from **877 to 288 ms**.

Prompt Memory — Chrome extension for prompt rewriting with approved memory — 2026
Co-built · source · evaluation corpus

- In-chat prompt rewriting on ChatGPT, Claude, Gemini, Perplexity and Grok (Manifest V3, FastAPI, Qdrant, MongoDB); context memory is written only after the user applies a rewrite.
- Built an 82-case adversarial evaluation corpus (21 failure categories) and revised the rewrite policy for source fidelity after it caught invented deadlines and limits.

SKILLS

Retrieval & LLMs	BGE-M3, Qdrant, SQLite FTS5, reciprocal rank fusion, cross-encoder reranking, RAG, LangGraph, LangChain, LoRA/QLoRA, evaluation design (MRR, nDCG)
ML & data	PyTorch, Transformers, scikit-learn, LightGBM, sentence-transformers, PySpark, SQL, Palantir Foundry
Engineering	Python, FastAPI, HTMX, Docker, SQLite/Turso, PostgreSQL, MongoDB, GitHub Actions, AWS, GCP

EDUCATION

Vellore Institute of Technology — B.Tech, CSE (Data Science) · CGPA 9.07 / 10 — May 2026
AWS Certified Cloud Practitioner · Machine Learning Specialization · IBM Data Science Professional Certificate